

README

1. Dataset title

Dataset belonging to Relationship Between Academic Stress, Mental Fatigue and Training Load in Adolescent Soccer Players

2. Researchers/Authors

- Names: Nijland, Rick^{1,2,3} (ORCID: 0000-0003-2442-6092); Toering, Tynke¹ (ORCID:0000-0003-2411-2755); Ivarsson, Andreas^{4,5} (ORCID: 0000-0002-8987-5975); de Jong, Johan^{1,2,3} (ORCID: 0000-0001-8300-3861); Lemmink, Koen^{2,3} (ORCID:0000-0001-8300-3861)
- Affiliation: ¹Hanze University of Applied Sciences; ²University Medical Center Groningen; ³University of Groningen; ⁴Halmstad University; ⁵University of Agder
- Contact: Use email button in DataverseNL to contact the corresponding researcher

3. Dataset Description

This dataset was collected as part of the research “The relationship between academic stress, mental fatigue and training load in adolescent soccer players”. This record contains raw data files, scripts to tidy and transform the data and scripts used for statistical analysis. This dataset was produced as part of the research project that aims to investigate the relationship between academic stress, mental fatigue and training load in adolescent soccer players. Using a prospective cohort design, 35 players (age 16.1 ± 1.2 years; length 176.0 ± 8.8 cm; body mass 64.4 ± 9.9 kg) were monitored for the second half of the 2022-2023 season. Players reported academic stress, mental fatigue and rating of perceived exertion on a daily basis. In addition, GPS and heart rate data were collected during training sessions using Polar Team Pro. After multiple imputation, multilevel analysis was conducted to predict internal load, i.e. Rate of Perceived Exertion (RPE) and Training Impulse (TRIMP). The dataset contains the anonymized data from the training sessions, as well as the accompanying codebooks, documentation and scripts. As the dataset does not contain sensitive information, it can be used freely.

Keywords: Adolescents (MeSH term);Mental fatigue (MeSH term); Academic stress; Perceived exertion; Training load; Exercise intensity; Athlete monitoring; Athlete wellness; High performance sports; Soccer; Multilevel analysis

4. Contents of this data record

File	Description
README.pdf	This document
Codebook_Relation_MF_AS_TL_dataverseNL.xlsx	Codebook belonging to this dataset. Contains codes for all scripts and file in the data.zip file
data_polar.zip	Zip file containing raw data files from Polar Team Pro organised per player
data.zip	Zip file containing raw data files from the questionnaires and additional information needed for analysis (e.g. times when teams trained, position of each player, maximal heart rate of each player, etc). Data files are all csv files from Microsoft Excel. Description of each file is embedded in the codebook (Codebook_Relation_MF_AS_TL_dataverseNL) because each file is imported in R as a data frame.
01_TL_calculation_v2_june2025.Rmd	Script used to import each Polar Team Pro file and process it to get the relevant study variables
02_Combine_DF_all_data_v2_june2025.Rmd	Script used to combine raw data from several sources and combine them into one data frame
03_data_tidying_and_visualisation_june2025.Rmd	Script used to tidy the data
04_Model_Building_v2_June2025.qmd	Script used for statistical analysis, including descriptives, missing data analysis, multiple imputation and multilevel modelling
04_Model_Building_v2_June2025.html	HTML file rendered from 04_Model_Building_v2_June2025.qmd

5. Data formats and structures

- **File format:**
 - .csv: Raw and processed data files (comma-separated)
 - .zip: Bundled datasets (e.g., grouped by player or source)
 - .pdf: Documentation (e.g., this README)
 - .Rmd: R Markdown scripts for data processing
 - .qmd: Quarto documents used for analysis and reporting
- **delimiter:** Comma (',')
- **Coding:** UTF-8
- **Missing values:** Represented as NA

6. Collection method

- **Method:** Observations
- **Location:** The Netherlands
- **Time span:** 23-01-2023 - 23-06-2023
- **Sample:** 35

7. Variables and description

See codebook [Codebook_Relation_MF_AS_TL_dataverseNL.xlsx] for more information on the variables of this dataset

8. Cleaning and processing

Raw data from Polar Team Pro was imported in R and processed to calculate training load variables using the script ‘01_TL_calculation_v2_june2025.Rmd’. Next step was to combine data from multiple sources in 1 data frame with the script ‘02_Combine_DF_all_data_v2_june2025.Rmd’. Thereafter, the data was cleaned and tidied using the next script ‘03_data_tidying_and_visualisation_june2025.Rmd’. In this script data was cleaned using several steps which are described in the script. For example, impossible values were removed (e.g. 5 for RPE), duplicates were removed and data from polar files with more than 5% of missing data were removed. Missing values were coded as NA with around 30% of the data missing. Visual and descriptive diagnostics were used to explore missingness patterns. Multiple imputation was performed using the mice package (predictive mean matching and 2-level models), where applicable. Sensitivity analysis was used for comparison.

9. Licenses and terms of use

- **License:** CC-BY-SA
- **Gebruiksvoorwaarden:** This data may be used in accordance with the license mentioned above. Upon reuse of this data, please reference this data set as follows: **Example citation:** Nijland, Rick; Toering, Tynke; Ivarsson, Andreas; de Jong, Johan; Lemmink, Koen (2025). “Dataset belonging to Relationship Between Academic Stress, Mental Fatigue and Training Load in Adolescent Soccer Players” [Data set]. DataverseNL.

10. Related publications

See metadata of this publication

11. Software and reproducibility

- **Software necessary:** [R]
- **Scripts available:** yes
- **Versions:** [R 4.3.1]

12. Known limitations

Limited to players belonging to two teams from one professional football club in the Netherlands. Some key variables (e.g., perceived exertion, academic stress, mental fatigue) rely on self-report questionnaires, which may be influenced by recall bias, social desirability, or inconsistent interpretation by players. Moreover, around 30% of missing data was present in the data set.

13. Last Update

- **Date:** 10-07-2025
- **Version:** v1.0